Machine Learning – CDS201

Module 2: Regression Models

Regression Models Evaluation Metrics

- The **sum of squares total (SST)** or the **total sum of squares (TSS)** is the sum of squared differences between the observed *dependent variables* and the overall **mean**.

$$SST = \sum_{i=1}^{n} (y_i - \bar{y})^2$$

- The **sum of squares due to regression (SSR)** or **explained sum of squares (ESS)** is the sum of the differences between the *predicted value* and the **mean** of the *dependent variable*. In other words, it describes how well our line fits the data.

$$SSR = \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2$$

- If **SSR** equals **SST**, our **regression model** perfectly captures all the observed variability, but that's rarely the case.

- The **sum of squares error (SSE) or residual sum of squares (RSS, where** residual means remaining or unexplained**)** is the difference between the *observed* and *predicted* values.

$$SSE = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

SST = SSE (unexplained) + SSR (explained)

- Regression analysis aims to minimize the SSE—the smaller the error, the better the regression's estimation power.

- $R^2 = SSR/SST$
- R^2 -> variability explained by the **regression**, divided by total variability.
- It is a relative measure and takes values ranging from 0 to 1.
- An **R-squared** of zero means our **regression line** explains none of the variability of the data.
- An **R-squared** of 1 would mean our model explains the entire variability of the data.

- R-squared is calculated mathematically by comparing the Sum of Squares of Errors (SSE) or the Sum of Squared Residuals (SSR) to the Total Sum of Squares (SST).

$$R^2 = 1 – (SSE/SST)$$

Analysis of Variance (ANOVA)

- ANOVA is a statistical procedure for analyzing the total variability of the dependent variable.
- The ANOVA table allows us to determine how fitting the regression model is to the observed data, which is a factor in how confident we are of the estimations given by the model.
- Total degrees of freedom: **n – 1**, where n is the number of observations.
- Degrees of freedom for the regression: **k**, which is the number of parameters (1 for simple linear regression).
- Degrees of freedom for the error: **n – k – 1** (for a simple linear regression, n – 2).

- **Mean of the Regression Sum of Squares (MSR):** MSR = RSS / k. For a simple linear regression, the MSR is simply equal to the RSS.

- **Mean of the Sum of Squared Errors (MSE):** MSE = SSE / (n – 2).

Anova table

ANOVA

Table

Source of Variation	Sum of Squares	In general Degrees of freedom (df)	Sum of Sq / df Mean Sum of Squares
Unexplained	SSE	$n-k-1 = n-2$ (for simple linear regression)	Mean Squared Error $MSE = SSE/(n-2)$
Explained	RSS	$k = 1$ (for simple linear regression)	Mean Regression Sum of Squares $MSR = RSS/1$
Total Variation	SST	$n - 1$	

n: number of observations ***k***: number of parameters

Goodness fit

- Standard Error of Estimate (SEE) : **standard deviation of the error term**, which is equal to the square root of the Mean Squared Error (MSE). This is called the **Standard Error of Estimate (SEE)**.

- **Interpreting SEE and R-squared:**
- **Low SEE and high R-squared** indicate a good fit, meaning we should have a higher confidence in the estimates.
- **High SEE and low R-squared** indicate a poor fit, and our confidence in the estimates should be low.